

Limitations of the Safe Drinking Water Act

Addressing the need for rural, isolated communities in the Navajo Nation

The Safe Drinking Water Act of 1974 was made to address the need for safe drinking water for the public in the United States. Public water is defined *as a system that provides water for human consumption to 15 or more connections or regularly serves 25 or more people daily for at least 60 days out of the year*. The definition breaks that down into more for more categories:

Community Water System (CWS)- a public water system that serves at least 15 service connections used by year-round residents or regularly serves at least 25 year-round residents.

Non-Transient, Non-Community Water Systems (NTNCWS)- a public water system that is not a “community water system: and that regularly serves at least 25 of the same persons for more than six months per year, including but not limited to schools, factories, and public buildings.

Transient, Non-Community Water System (TNCWS)- A public water system that is neither a “community water system” nor a “community water system,” including but limited to seasonal facilities such as children’s camps or recreational camping areas: and year-round facilities that serve more than 25 persons who are not residents thereof, such as a gasoline service stations, marinas, rest areas and restaurants that are not operated by a community water system. These systems must sample bacteriological quality quarterly and nitrate/nitrite annually.

Consecutive Public Water System- A public water system purchases water from a primary purveyor. This type of water system typically purchases water, pumps it into its storage tanks, and distributes it through pipes (i.e., schools, stores, etc.). These systems must monitor for bacteriological quality every month, lead/copper, and asbestos (if asbestos-cement (AC) pipes are used in the distribution).

Public Water Systems (navajopublicwater.org)

Public water, by definition and breakdown of each of the four categories, shows the limitation of the Safe Drinking Water Act on the Navajo Nation. In some cases, the number of people that rely on unregulated water goes from 30% to 40% in different areas of the Navajo Nation. In the general U.S. population, that number is 3% (I NEED TO LOOK THIS UP AGAIN FOR CONFIRMATION). So that means we need to rethink how more people in the Navajo Nation's isolated rural areas can access potable water that does not jeopardize their health.

Grassroots Organizations

The grassroots organizations can help fill the void of the Safe Drinking Water Act (SDWA), where the regulator cannot provide the need for safe potable water. The tribal and federal regulators follow the Safe Drinking Water Act (SDWA), where population density is the driving factor.

This has the western concept of providing water to the masses but does not satisfy tribal needs and does not help preserve tribal culture. The Office of Environmental Health under the Indian Health Services extends their public water to areas that meet the needs for potable water as long as the requirements are met for the SDWA. This excludes tribal members that live in rural, isolated areas that are preserving their cultural knowledge. This is where the grassroots organization can close the gap on the lack of water infrastructure. The grassroots are local community members helping improve the quality of life for their community. They have community connections that surpass the tribal and federal regulators in terms of the trust.

Well improvement

Wells that are considered unregulated wells can be improved for communities that rely on the wells. The same can be done to springs, windmills, and hand pump wells. Some of these locations are used for cultural practice in ceremonies. Therefore, improving them would help preserve the teaching traditions around water.

Many of the springs accessible to the community members are local knowledge that the community members rely on. This has been a safety net for many community members when the water infrastructure has been stressed to the point where the regulated watering point is turned off.

Under the SDWA, the springs, wells, or windmills are, therefore, a community water system that the people utilize for their water needs. The grassroots organizations can work on these springs with the help of academic institutions.

Water Filtration System

Some wells, springs, handpump wells, and windmills are high in heavy metals such as uranium and arsenic. These same wells can be made safe for human consumption by providing a water filtration system.

They are grassroots organizations addressing the issue starting at the local level and identifying the wells, analyzing the water quality, working with academic institutions, and conducting

research to address water contamination better using best practices. This helps empower the community to be more self-reliant.

Water Cost and Analysis

The grassroots organizations are working with certified labs. The community members will be trained to sample Community Water System (CWS) to comply with the SDWA. The grassroots organization will pay for the analysis and share the data with the Public Water System Supervision Program to regulate the CWS and comply with federal and tribal policy.